MODULE 57 · DOMAIN 5 - OPERATIONAL EXCELLENCE AND AUTOMATION

Practice Exam: Domain 5

Fifteen original questions on operational excellence and automation. Pause, decide, then watch.

DOMAIN 5 of 5 EXAM WEIGHT 20% SET 5 of 6

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How to use this set

- **Allow 2.4 minutes per question**: 36 minutes for the set.
- **Write your answer down** before you unpause.
- **Ask when the control acts**: before, during or after.
- Custom glue that recreates a built-in feature is usually a distractor.

> ON THE EXAM
>
> Timing words decide most operations questions: before deployment, before users notice, before the load.

A platform team must deploy a standard set of IAM roles and Config rules to every account in two OUs. Accounts are added to these OUs every week.

New accounts must receive the stack automatically, with no manual steps.

What should the solutions architect use?

A CloudFormation StackSets with self-managed permissions, updated weekly

B CloudFormation StackSets with service-managed permissions and automatic deployment to the OUs

C A Lambda function that deploys stacks when accounts are created

D A separate CloudFormation stack deployed by hand in each account

◆ PAUSE AND DECIDE

What should the solutions architect use?

A StackSets, self-managed
Needs roles in each account and manual targeting.

B StackSets, service-managed
Deploys to new OU accounts automatically.

C Custom Lambda
Custom glue for a built-in feature.

D Manual stacks
Manual steps every week.

THE SIGNAL

Every account in an OU, including new ones, with no manual steps: service-managed with auto-deployment. Module 40.

> An engineer changed an ingress rule on a security group in the console. The security group belongs to a CloudFormation stack, and the template is correct.
>
> The team wants to restore the stack's resources to match the template, previewing the changes first.

What should the solutions architect do?

A Run drift detection, then create a drift-aware change set that reverts the drift, review it, and execute it

B Delete and recreate the stack

C Edit the security group back by hand

D Add an AWS Config rule

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Drift-aware change set
Previews and restores template values.

B Recreate stack
Outage and needless risk.

C Manual fix
Error-prone; no preview or record.

D Config rule
Detects; does not restore the template.

THE SIGNAL

A stack resource changed in the console: drift-aware change set. Module 40.

> A company's rule says every S3 bucket must block public access. Some teams deploy with CloudFormation, others with Terraform using the AWS Cloud Control provider.
>
> Non-compliant buckets must be rejected before they are created.

What should the solutions architect implement?

A A CDK aspect in a shared library

B A weekly report from AWS Config

C Amazon Macie

D A CloudFormation Guard Hook in fail mode, activated in every account

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A CDK aspect
Only covers CDK apps.

B Weekly Config report
After creation, and slow.

C Macie
Finds sensitive data; off the in-scope list.

D Guard Hook, fail mode
Rejects before provisioning, for both tools.

THE SIGNAL

Block before provisioning, CloudFormation and Cloud Control users alike: a Guard Hook. Module 41.

> An audit found EBS volumes without encryption in 30 accounts. Most were created years ago through the console and the CLI.
>
> Security wants all such volumes found across accounts, and wants future violations reported and remediated automatically.

What should the solutions architect implement?

A. A stricter CloudFormation Hook

B. A drift-aware change set on every stack

C. AWS Config rules deployed organisation-wide, with automatic remediation through Systems Manager Automation runbooks

D. A CDK aspect

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Stricter Hook
Never sees console or CLI changes.

B Drift-aware change sets
Only for stack resources.

C Config with remediation
Finds every resource, however created, and remediates.

D CDK aspect
Only for CDK apps.

THE SIGNAL

Created by hand, found and fixed: Config rules with remediation. Module 41.

> A pipeline in a tooling account deploys CloudFormation stacks to a production account. The artefact bucket uses SSE-KMS with the AWS managed key.
>
> The production deploy action fails with access denied when reading the artefact.

What should the solutions architect change?

A. Encrypt artefacts with a customer managed KMS key whose key policy, and the bucket policy, allow the production deployment role

B. Disable encryption on the artefact bucket

C. Add the production account to the AWS managed key's policy

D. Run a separate pipeline inside the production account

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Customer managed key
Its policy can allow another account.

B No encryption
Weakens security needlessly.

C Edit the AWS managed key
AWS managed key policies cannot be edited.

D Pipeline per account
Abandons build once, promote.

THE SIGNAL

Cross-account artefacts: a customer managed KMS key shared through its key policy. Module 42.

A team keeps eight services in one GitHub repository, each in its own folder with its own V2 pipeline. Every commit starts all eight pipelines.

Each pipeline should start only when files in its own folder change.

What should the solutions architect configure?

A. A Lambda function that inspects commits and starts pipelines

B. Separate repositories for each service

C. Trigger filters on each pipeline with a file path include pattern for its folder

D. Polling on a five-minute schedule

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Lambda dispatcher
Custom code for a built-in feature.

B Split repositories
Major workflow change.

C File path trigger filters
Native per-folder triggering.

D Polling
Still triggers on any change.

THE SIGNAL

Monorepo, start only the affected pipeline: trigger file path filters. Module 42.

> A company wants container images with critical vulnerabilities stopped in CodeBuild, before they are ever pushed to Amazon ECR.
>
> The check must use Amazon Inspector's vulnerability data.

What should the solutions architect add to the build?

A. Rely on ECR enhanced scanning after push

B. Generate an SBOM with the Inspector SBOM generator, scan it with the Inspector ScanSbom API, and fail on critical findings

C. Enable GuardDuty Runtime Monitoring

D. Add an AWS Config rule

◆ PAUSE AND DECIDE

What should the solutions architect add to the build?

A Enhanced scanning
Runs after the image is in ECR.

B SBOM scan in CodeBuild
Before push, with Inspector's data.

C GuardDuty Runtime
Detects runtime threats, not CVEs.

D Config rule
Checks configuration, not images.

THE SIGNAL

Stop it before the registry, with Inspector's data: an SBOM scan in the build. Module 43.

A bank's EKS clusters must run only images built by its own pipeline. Last month, a pod ran an image pulled directly from a public registry.

Security wants a cryptographic guarantee, with signing keys managed by AWS.

What should the solutions architect implement?

A ECR tag immutability

B IAM policies limiting who can push to ECR

C ECR enhanced scanning

D ECR managed signing with an AWS Signer profile, and an admission controller that verifies signatures

◆ PAUSE AND DECIDE

What should the solutions architect implement?

A Tag immutability
Does not stop foreign images.

B Push permissions
Public images bypass ECR.

C Enhanced scanning
Finds CVEs, not provenance.

D Sign and verify
Unsigned images refused at admission.

THE SIGNAL

Only our images may run, cryptographically: sign on push, verify at admission. Module 43.

A trading firm deploys a pricing Lambda function. A new version must take 10 percent of traffic for 10 minutes, then all of it.

If the error alarm fires during that time, traffic must return to the previous version automatically.

What should the solutions architect use?

A CodeDeploy with a canary 10 percent, 10 minute configuration on the function alias, with the alarm attached

B Route 53 weighted records for two function URLs

C Publish the new version and update callers

D An API Gateway stage variable switched by hand

◆ PAUSE AND DECIDE

What should the solutions architect use?

A CodeDeploy canary on alias
Shifts alias weight, rolls back on alarm.

B Route 53 weights
DNS caching; no automatic rollback.

C Update callers
All traffic at once.

D Manual stage switch
Manual, no rollback.

THE SIGNAL

Lambda, small slice first, automatic revert: CodeDeploy canary on the alias. Module 44.

A logistics company uses ECS rolling updates. A release with a bad configuration caused new tasks to fail health checks repeatedly.

The deployment kept retrying for hours until someone noticed and rolled back by hand.

What should the solutions architect enable?

A. A longer health check grace period

B. Blue/green deployments with double capacity

C. A higher minimum healthy percent

D. The ECS deployment circuit breaker with rollback enabled

◆ PAUSE AND DECIDE

What should the solutions architect enable?

A Longer grace period
Delays detection.

B Blue/green
Larger change than needed.

C Higher minimum healthy
Protects capacity, not retries.

D Circuit breaker
Stops failing rollouts and rolls back.

THE SIGNAL

ECS tasks keep failing during a rolling update: the deployment circuit breaker. Module 44.

A retailer is adding distributed tracing to 50 services on ECS and Lambda. A developer proposes the X-Ray SDK and daemon from an older project.

The company wants instrumentation it will not have to replace, and may send traces to a second tool later.

What should the solutions architect recommend?

A. The X-Ray SDK and daemon

B. VPC Flow Logs

C. OpenTelemetry instrumentation through the AWS Distro for OpenTelemetry

D. CloudTrail data events

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A X-Ray SDK
Maintenance mode; end of support February 2027.

B Flow Logs
Network records, not traces.

C ADOT and OpenTelemetry
Supported, vendor-neutral, exports to X-Ray.

D CloudTrail
API audit, not request traces.

THE SIGNAL

New instrumentation: OpenTelemetry through ADOT, not the X-Ray SDK. Module 45.

> A travel site's synthetic canaries in eu-west-1 report fast page loads. Customers in Australia and Brazil complain that pages are slow on mobile.
>
> The team wants actual load times and JavaScript errors by country, browser and device.

What should the solutions architect add?

A More canaries in eu-west-1

B CloudWatch RUM, with its web client added to the site

C X-Ray tracing on the back end

D Application Load Balancer access logs

◆ PAUSE AND DECIDE

What should the solutions architect add?

A More canaries
Same Region; not real users.

B CloudWatch RUM
Real sessions by country, browser and device.

C X-Ray
Back-end timing only.

D ALB logs
Server timing; no browser data.

THE SIGNAL

What real users see, by place and device: RUM. Before users notice, at quiet times: Synthetics. Module 46.

A partner API gets about 20 requests per minute overnight. Its availability SLO is period-based, so a single failed request marks a whole minute unhealthy.

Overnight attainment looks far worse than the true share of failed requests.

What should the solutions architect change?

A Use a five-minute period

B Exclude overnight hours from the SLO

C Switch to a latency SLI

D Use a request-based SLI, counting good requests against total requests

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Longer period

One failure now spoils five minutes.

B Exclude nights

Hides real failures.

C Latency SLI

Measures something else.

D Request-based SLI

Attainment matches the real failure ratio.

THE SIGNAL

Low or uneven traffic, period health misleading: a request-based SLI. Module 46.

> Five applications share the same Bedrock model in one account. The model's cost has doubled, and finance wants to know which application caused it and which prompts are largest.
>
> Only aggregate token metrics are available today.

What should the solutions architect do?

A. Give each application a tagged application inference profile, and enable model invocation logging

B. Read CloudTrail management events

C. Enable X-Ray tracing on the model

D. Divide the bill evenly

◆ PAUSE AND DECIDE

What should the solutions architect do?

A Inference profiles and invocation logs
Cost per application; prompts per request.

B CloudTrail
No token counts.

C X-Ray
Timing, not token attribution.

D Even split
Guesswork.

THE SIGNAL

Which application and which prompts drive spend: inference profiles and invocation logs. Module 47.

> An application on EC2 behind an Application Load Balancer sometimes deadlocks. The instance passes EC2 status checks but stops answering, and the load balancer marks it unhealthy.
>
> The Auto Scaling group never replaces it.

What should the solutions architect change?

A Enable EC2 automatic recovery

B Reboot instances on high CPU with a CloudWatch alarm

C Set the Auto Scaling group health check type to ELB, with a suitable grace period

D Use a larger instance type

◆ PAUSE AND DECIDE

What should the solutions architect change?

A Auto recovery

Hardware and system failures only.

B Reboot on CPU

A deadlock may not raise CPU.

C ELB health checks

Group replaces what the ALB finds unhealthy.

D Larger instance

Deadlocks are not about size.

THE SIGNAL

Load balancer sees it unhealthy, group does not replace it: ELB health check type. Module 48.

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Where to revisit

IF YOU MISSED	REWATCH
Q1, Q2	Module 40: Infrastructure as code
Q3, Q4	Module 41: Policy as code and drift
Q5, Q6	Module 42: CI/CD pipeline design
Q7, Q8	Module 43: DevSecOps and AI evaluation
Q9, Q10	Module 44: Deployment strategies
Q11	Module 45: Observability foundations
Q12, Q13	Module 46: User experience and SLOs
Q14	Module 47: AI observability
Q15	Module 48: Automation and resilience

ON THE EXAM

12 or more out of 15 is on track. Next, the full 75-question mock in three parts.